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Studierichting: Informatica

Oefeningenreeks 5J nr. 2.4

$$\begin{cases} x(t) = a \cos^3(t) \\ y(t) = a \sin^3(t) \end{cases}$$

$$x'(t) = -3a \cos^2(t) \sin(t)$$

$$y'(t) = 3a \sin^2(t) \cos(t)$$

$$S = \int_0^{2\pi} \sqrt{9a^2 \cos^4(t) \sin^2(t) + 9a^2 \cos^2(t) \sin^4(t)} dt$$

$$= 3a \int_0^{2\pi} \sqrt{\cos^2(t) \sin^2(t) (\cos^2(t) + \sin^2(t))} dt$$

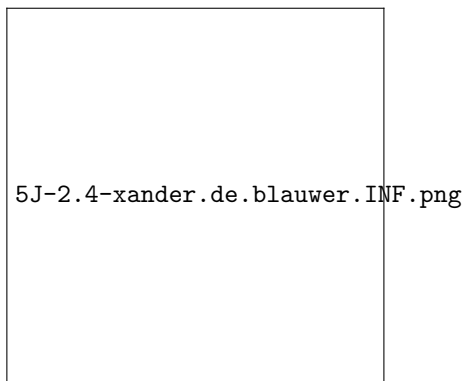
$$= 3a \int_0^{2\pi} \cos(t) \sin(t) dt$$

$$= 4 \cdot 3a \int_0^{\frac{\pi}{2}} \cos(t) \sin(t) dt$$

$$\text{Stel } x = \sin(t) \Rightarrow dx = \cos(t) dt$$

$$= \frac{4 \cdot 3}{2} a [\sin^2(t)]_0^{\frac{\pi}{2}}$$

$$= 6a$$



References